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k-Means, k-Medoids

k-Means Example, Hierarchical Clustering

Gaussian Mixture Model

Video: What is the Gaussian Mixture Model? 13 min

Reading: Mixture Models 1h

Practice Assignment: Quiz 12 Submitted

Graded Assignment: Final Test Submitted

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Final Test

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

1. What is the 10 fold cross validation error for maximum margin separation in the following figure? **1 / 1 point**

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0

Correct

Try again

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2. Find the answer using conditional probability under the following conditions. **1 / 1 point**

- 50% of the mail I received was spam
- 95% of spam mail had the word 'cash'
- 2% of normal emails had the word 'cash'

If I got one e-mail right now, and the e-mail contains the word 'cash',find out the probabilitythat this is a spam e-mail.

- (•) 95/97
- ( ) 98/100
- ( ) 97/100
- ( ) 95/100
- ( ) 95/98

Correct

3. Choose all that are not correct in the description of the decision tree. (10 points) **1 / 1 point**

- (•) Entropy means disorder, and the higher this number, the more ordered.
- ( ) Decision is a model that makes rules in the form of a tree.
- ( ) The evaluation of whether the decision was made well can be made even with entropy.
- ( ) (0,0,1,1) is more chaotic than (0,0,0,1).
- ( ) Decision tree builds a model through the process of lowering entropy.

Correct

4. The following dataset will be used to learn a decision tree for predicting whether a person is happy (H) or sad (S) based on the color of their shoes, whether they wear a wig and the number of ears they have. **1 / 1 point**

Color	Wig	Num. Ears	(Output) Emotion
G	Y	2	S
G	N	2	S
G	N	2	S
B	N	2	S
B	N	2	H
R	N	2	H
R	N	2	H
R	N	2	H
R	Y	3	H

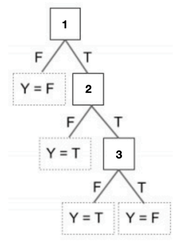
1) What is P(Emotion=H | Wig=Y)?
2) What is P(Emotion=H | # of Ears=3)?

- (•) 0.5, 1
- ( ) 1, 0.5
- ( ) 0.25, 1
- ( ) 0.25, 0.5

Correct

5. Using the dataset bellow, we want to build a decision tree which classifies Y as True/False given the binary variables A, B, C. Draw the tree that would be learned by the greedy algorithm with zero training error. You do not need to show any computation. **1 / 1 point**

A	B	C	Y
False	False	False	False
True	False	True	True
True	True	False	True
True	True	True	False

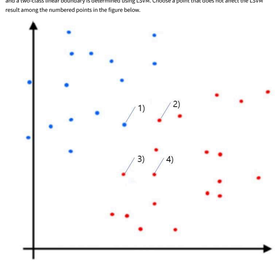


Answer with format by 1-2-3
ex) D-E-F

A-B-C

Correct

6. As shown below, the data set is displayed on a two-dimensional graph according to two labeling (Red/Blue), and a two-class linear boundary is determined using LSVM. Choose a point that does not affect the LSVM result among the numbered points in the figure below. **1 / 1 point**



- ( ) 1
- ( ) 2
- ( ) 3
- (•) 4

Correct

7. Which of the following statements about C (Cost) in the SVM model is incorrect? **1 / 1 point**

- ( ) Hard margin SVM and soft margin SVM can be distinguished.
- (•) It is not suitable for determining the weight for the error.
- ( ) If C is too small, under-fitting is highly likely.
- ( ) It is designed to solve the case where the data cannot be completely separated linearly.
- ( ) c determines the degree to which data samples are allowed to be placed in different classes

Correct

8. What are the advantages and disadvantages of random forests? **1 / 1 point**

- ( ) Ability to handle data without preprocessing.
- ( ) Data does not need to be rescaled, transformed, or modified Resistant to outliers.
- ( ) It can handle thousands of input variables without variable deletion.
- (•) It is less effective for training and has more parameters than SVMs ans NNs.

Correct

9. Which of the following is the correct order for Gaussian Mixture Model? (10 points) **1 / 1 point**

A) Evaluate for all i and j
B) Randomly initialize
C) Go back to step2, stop until parameters don't change
D) Evaluate for all i

Answer with format X-X-X-X

ex) E-F-G-H

B-A-D-C

Correct

10. Which of the following statements about cluster and clustering is not correct? **1 / 1 point**

- ( ) A cluster means a group of the similar objects.
- ( ) A good cluster has high intra-cluster similarity.
- (•) Clustering has pre-determined group information.
- ( ) A good cluster has a high inter-cluster distance.

Correct